Élie de Panafieu

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RESEARCH INTERESTS

Analytic combinatorics; enumeration and structure of random graphs, digraphs and hypergraphs; analysis of algorithms; computational social choice and voting systems; random instances of constraint satisfaction problems.

EMPLOYMENT

2016–present	**Researcher**, Nokia Bell Labs France, Paris-Saclay Centre. Permanent research position since January 2016.
2015	**Postdoctoral researcher** (September–December), LIP6, Sorbonne Université, *Complex Networks* team, with Clémence Magnien and Matthieu Latapy.
2014–2015	**Postdoctoral researcher**, Research Institute for Symbolic Computation (RISC), Johannes Kepler Universität Linz, Austria, with Manuel Kauers.

PUBLICATIONS

Summary. 11 articles in international peer-reviewed journals, 16 papers in international peer-reviewed conferences, 2 papers in French national conferences, 2 preprints under review, 1 PhD thesis and 2 patent applications, written with 38 distinct co-authors.

A complete list is given at the end of this CV.

EDUCATION

2011–2014	**PhD in computer science**, Université Paris-Diderot (Sorbonne Paris-Cité), LIAFA, supervised by Vlady Ravelomanana. Thesis: *Analytic Combinatorics of Graphs, Hypergraphs and Inhomogeneous Graphs.*
2010	**Agrégation externe de mathématiques**, computer science option. One-year research internship in computer algebra, *Algorithms* team, MSR–INRIA Joint Centre, with Alin Bostan and Frédéric Chyzak.
2009	**Master 2 (MPRI)**, Master Parisien de Recherche en Informatique, with distinction (mention *Bien*). Five-month research internship in the same *Algorithms* team.
2008	**Master 1 (MPRI)**; joint Licence 3 and Master 1 in mathematics, ENS Paris-Saclay (formerly ENS Cachan) and Université Paris-Cité (formerly Paris-Diderot). Three-month internship in cryptography, Université catholique de Louvain (Louvain-la-Neuve, Belgium), with Benoît Libert.
2007	**Licence 3 in computer science**, ENS Paris-Saclay; admission to ENS Paris-Saclay. Three-month internship in cryptography, LORIA (Nancy), *CACAO* team, with Emmanuel Thomé and Paul Zimmermann.
2004–2007	Classes préparatoires aux grandes écoles, Lycée Charlemagne, Paris.

RESPONSIBILITIES AND SCIENTIFIC ANIMATION

RandNET	Local coordinator at Nokia Bell Labs for RandNET (*Randomness and Learning in Networks*), a Marie Skłodowska-Curie Actions Research and Innovation Staff Exchange project (MSCA-RISE) involving 14 research institutions, 2021–2026.

	Coordinated by Marc Noy (Universitat Politècnica de Catalunya), the project brought together researchers in combinatorics, probability theory and computer science to study large graph-like objects and their applications.
ANR PROSOC	Main coordinator of PROSOC (*Probability and Social Choice*), an ANR project submitted to the AAPG 2026 call, ranked first on the waiting list. A resubmission is in preparation. The project brings together 26 researchers in combinatorics, analysis of algorithms and computational social choice from Nokia Bell Labs, LIPN (Laboratoire d'Informatique de Paris Nord) and LIP6 (Laboratoire d'Informatique de Sorbonne Université). Its aim is to study the distribution of the complexity of social choice algorithms, such as voting rules, rather than their worst case, with an emphasis on the interaction between artificial intelligence and computational social choice.
ANR PandaG	Regular invited participant in the meetings of the ANR project PandaG (*Analyse de paramètres de classes de DAGs*). Started in 2023 and coordinated by Antoine Genitrini, the project brings together researchers from GREYC (Groupe de Recherche en Informatique, Image, Automatique et Instrumentation de Caen), LIP6 and LIPN to study parameters of various classes of directed acyclic graphs.
LINCS	Associate member of LINCS (Laboratory for Information, Networking and Communication Sciences), 2016–2025. LINCS is a joint academic-industrial laboratory working on future information and communication networks, systems and services. Its partners are Inria, Institut Mines-Télécom, Nokia Bell Labs, Sorbonne Université and SystemX.

RESEARCH SUPERVISION

PhD students

2022–2025	**Emma Caizergues** — *Analytic Combinatorics and the Probability of the Condorcet Paradox*. Université PSL, prepared at Université Paris-Dauphine PSL, doctoral school ED SDOSE (no. 543). Defended 18 December 2025. Thesis director: Jérôme Lang (CNRS, Université Paris-Dauphine PSL). Co-supervisors: François Durand and Élie de Panafieu (Nokia Bell Labs). Jury: Frédérique Bassino (Université Sorbonne Paris Nord, president), Vincent Jugé (Université Gustave Eiffel, referee), Antonin Macé (CNRS, Paris School of Economics, referee), Vincent Merlin (CNRS, Université de Caen-Normandie, examiner), Marc Noy (Universitat Politècnica de Catalunya, examiner).
2019–2022	**Quentin Lutz** — *Graph-based Contributions to Machine Learning*. Institut Polytechnique de Paris, prepared at Télécom Paris, doctoral school EDIPP (no. 626), specialty Computer Science, Data and Artificial Intelligence. Defended 9 February 2022. Thesis director: Thomas Bonald (Télécom Paris). Co-supervisor: Élie de Panafieu (Nokia Bell Labs). Jury: Jean-Loup Guillaume (Université de La Rochelle, president), Matthieu Latapy (CNRS and Sorbonne Université, referee), Conrado Martínez (Universitat Politècnica de Catalunya, referee), Cécile Mailler (University of Bath, examiner), Gérard Burnside (Nokia Bell Labs, invited).
2022–2025	**Breno Skuk** — CIFRE PhD, Télécom Paris and Nokia Bell Labs France. *Neural network quantization techniques*. Nokia Bell Labs France supervision taken over in 2023. The thesis was not completed.

Students and interns

Jan 2025–Mar 2025	**Marta Teodora Trales** — bachelor thesis, École Polytechnique, Institut Polytechnique de Paris. *Grid Coloring for Object Sensing and Localization*. Co-supervised with Cédric Adjih (INRIA) and Pierre Escamilla (Nokia Bell Labs).

TEACHING

Regular teaching at master's level since 2019, alongside earlier undergraduate teaching at Université Paris-Cité and Sorbonne Université, and several years of secondary-school teaching.

Master's level

2022–present **Analytic Combinatorics** — M2, Master *Mathématiques de l'aléatoire*, Université Paris-Saclay. 12–21 h per year.

2019–present **Analysis of Algorithms** — M2, MPRI (Master Parisien de Recherche en Informatique), Université Paris-Cité. 12 h per year.

2017–2018 **Advanced Algorithms** — M1, Sorbonne Université. 28 h of lab sessions.

Undergraduate level

2013–2014 **Internet and Tools** — L1, Université Paris-Cité. 52 h of lab sessions. HTML, CSS, PHP, MongoDB.

2012–2013 **Discrete Mathematics** — L3, Université Paris-Cité. 26 h of exercise classes. Regular and algebraic languages, generating functions.

Introduction to Programming — L1, Université Paris-Cité. 39 h of lab sessions. Java.

2011–2012 **Introduction to Programming** — L1, Université Paris-Cité. 39 h of lab sessions and 26 h of lectures and exercise classes.

Secondary school and outreach

2021–2022 **Digital and Computer Science (NSI)** — Year 11, Lycée École Alsacienne. 60 h of lectures and exercise classes.

2019 One-week mathematics course for high-school students, Brazzaville (Republic of the Congo). Association Animath.

2018–2019 **Introduction to Computer Science** — Year 11, Lycée École Alsacienne. 30 h of lectures and exercise classes. Python.

2015–2017 **Preparation for the Sciences Po entrance examination in mathematics** — Years 11 and 12, Lycée École Alsacienne. 24 h.

2008–2010 **Introduction to Computer Science** — Years 11 and 12, Lycée École Alsacienne. 120 h of lectures and exercise classes. Caml.

LIST OF PUBLICATIONS

Conference papers in this field are peer-reviewed and count as full publications; where a conference paper was later extended into a journal article, both are listed.

Articles in international peer-reviewed journals

1. Hainzl, Eva-Maria, and Élie de Panafieu. 2026. "Tree Walks and the Spectrum of Random Graphs." *Eur. J. Comb.* 136:104386.. https://doi.org/10.1016/j.ejc.2026.104386
2. de Panafieu, Élie, and Michael Wallner. 2026. "Combinatorics of Nondeterministic Walks." *Comb. Theory*.. https://arxiv.org/abs/2311.03234 *Accepted; to appear.*
3. Caizergues, Emma, François Durand, Marc Noy, Élie de Panafieu, and Vlady Ravelomanana. 2026. "Probability of a Condorcet Winner for Large Electorates: An Analytic Combinatorics Approach." *J. Artif. Intell. Res.*. https://arxiv.org/abs/2505.06028 *Accepted; to appear.*
4. Dovgal, Sergey, Élie de Panafieu, Dimbinaina Ralaivaosaona, Vonjy Rasendrahasina, and Stephan Wagner. 2024. "The Birth of the Strong Components." *Random Struct. Algorithms* 64(2):170–266.. https://arxiv.org/abs/2009.12127
5. Dovgal, Sergey, Élie de Panafieu, and Vlady Ravelomanana. 2023. "Exact Enumeration of Satisfiable 2-SAT Formulae." *Comb. Theory* 3(2):38.. https://arxiv.org/abs/2108.08067
6. Collet, Gwendal, Élie de Panafieu, Danièle Gardy, Bernhard Gittenberger, and Vlady Ravelomanana. 2020. "Threshold Functions for Small Subgraphs in Simple Graphs and Multigraphs." *Eur. J. Comb.* 88:103113.. https://doi.org/10.1016/j.ejc.2020.103113
7. de Panafieu, Élie. 2019. "Analytic Combinatorics of Connected Graphs." *Random Struct. Algorithms* 55(2):427–95.. https://doi.org/10.1002/rsa.20836
8. de Panafieu, Élie, Danièle Gardy, Bernhard Gittenberger, and Markus Kuba. 2016. "2-Xor Revisited: Satisfiability and Probabilities of Functions." *Algorithmica* 76(4):1035–76.. https://doi.org/10.1007/s00453-016-0119-x
9. de Panafieu, Élie, and Vlady Ravelomanana. 2015. "Analytic Description of the Phase Transition of Inhomogeneous Multigraphs." *Eur. J. Comb.* 48:186–97.. https://arxiv.org/abs/1409.8424
10. de Panafieu, Élie. 2015. "Phase Transition of Random Non-Uniform Hypergraphs." *J. Discrete Algorithms* 31:26–39.. https://doi.org/10.1016/j.jda.2015.01.009

11. Attrapadung, Nuttapong, Javier Herranz, Fabien Laguillaumie, Benoît Libert, Élie de Panafieu, and Carla Ràfols. 2012. “Attribute-Based Encryption Schemes with Constant-Size Ciphertexts.” *Theor. Comput. Sci.* 422:15–38.. https://doi.org/10.1016/j.tcs.2011.12.004

Papers in international peer-reviewed conferences

1. Durand, François, Élie de Panafieu, and Guillem Perarnau. 2026. “Super Condorcet Winners and Limit Coalitional Manipulability of IRV.” in *Proceedings of the 35th International Joint Conference on Artificial Intelligence (IJCAI).*
2. de Panafieu, Élie, François Durand, and Jérôme Lang. 2026. “The Communication Complexity of Instant-Runoff Voting.” in *Proceedings of the 35th International Joint Conference on Artificial Intelligence (IJCAI).*
3. Hainzl, Eva-Maria, and Élie de Panafieu. 2024. “Tree Walks and the Spectrum of Random Graphs.” P. 11:1–11:15 in *Proceedings of the 35th International Conference on Probabilistic, Combinatorial and Asymptotic Methods for the Analysis of Algorithms (AofA).* Vol. 302. Wadern: Schloss Dagstuhl – Leibniz Zentrum für Informatik
4. Chen, Chung Shue, Peter Keevash, Sean Kennedy, Élie de Panafieu, and Adrian Vetta. 2024. “Robot Positioning Using Torus Packing for Multisets.” P. 43:1–43:18 in *Proceedings of the 51st International Colloquium on Automata, Languages and Programming (ICALP).* Vol. 297. Wadern: Schloss Dagstuhl – Leibniz Zentrum für Informatik
5. Caizergues, Emma, and Élie de Panafieu. 2023. “Exact Enumeration of Graphs and Bipartite Graphs with Degree Constraints (Extended Abstract).” Pp. 254–62 in *Proceedings of the 12th European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB).* Brno: Masaryk University
6. Lutz, Quentin, Élie de Panafieu, Maya Stein, and Alex Scott. 2021. “Active Clustering for Labeling Training Data.” Pp. 8469–80 in *Advances in Neural Information Processing Systems 34 (NeurIPS).* Vol. 34.
7. de Panafieu, Élie, and Sergey Dovgal. 2020. “Counting Directed Acyclic and Elementary Digraphs.” *Sémin. Lothar. Comb.* 84B:12.. https://arxiv.org/abs/2001.08659
8. de Panafieu, Élie, Mohamed Lamine Lamali, and Michael Wallner. 2019. “Combinatorics of Nondeterministic Walks of the Dyck and Motzkin Type.” Pp. 1–12 in *Proceedings of the 16th Workshop on Analytic Algorithmics and Combinatorics (ANALCO).* Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM)
9. Bouillard, Anne, Céline Comte, Élie de Panafieu, and Fabien Mathieu. 2018. “Of Kernels and Queues: When Network Calculus Meets Analytic Combinatorics.” Pp. 49–54 in *Proceedings of the 30th International Teletraffic Congress (ITC).* Vol. 2. IEEE
10. Collet, Gwendal, Élie de Panafieu, Danièle Gardy, Bernhard Gittenberger, and Vlady Ravelomanana. 2017. “Threshold Functions for Small Subgraphs: An Analytic Approach.” Pp. 271–77 in *Extended Abstracts of the 9th European Conference on Combinatorics, Graph Theory and Applications (EUROCOMB).* Vol. 61, *Electron. Notes Discrete Math.* Amsterdam: Elsevier
11. de Panafieu, Élie, and Lander Ramos. 2016. “Graphs with Degree Constraints.” Pp. 34–45 in *Proceedings of the 13th Workshop on Analytic Algorithmics and Combinatorics (ANALCO).* Philadelphia, PA: Society for Industrial and Applied Mathematics (SIAM)
12. de Panafieu, Élie. 2016. “Counting Connected Graphs with Large Excess.” Pp. 979–90 in *Proceedings of the 28th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC).* Nancy: The Association. Discrete Mathematics & Theoretical Computer Science (DMTCS)
13. de Panafieu, Élie. 2015. “Enumeration and Structure of Inhomogeneous Graphs.” Pp. 901–12 in *Proceedings of the 27th International Conference on Formal Power Series and Algebraic Combinatorics (FPSAC).* Nancy: The Association. Discrete Mathematics & Theoretical Computer Science (DMTCS)
14. de Panafieu, Élie, Danièle Gardy, Bernhard Gittenberger, and Markus Kuba. 2014. “Probabilities of 2-Xor Functions.” Pp. 454–65 in *Proceedings of the 11th Latin American Symposium on Theoretical Informatics (LATIN).* Vol. 8392, *Lecture Notes in Computer Science.* Berlin: Springer
15. Bostan, Alin, Frédéric Chyzak, and Élie de Panafieu. 2013. “Complexity Estimates for Two Uncoupling Algorithms.” Pp. 85–92 in *Proceedings of the 38th International Symposium on Symbolic and Algebraic Computation (ISSAC).* New York, NY: Association for Computing Machinery (ACM)
16. Attrapadung, Nuttapong, Benoît Libert, and Élie de Panafieu. 2011. “Expressive Key-Policy Attribute-Based Encryption with Constant-Size Ciphertexts.” Pp. 90–108 in *Proceedings of the 14th International Conference on Practice and Theory in Public Key Cryptography (PKC).* Vol. 6571, *Lecture Notes in Computer Science.* Berlin: Springer

Papers in French national conferences

1. de Panafieu, Élie, Mohamed Lamine Lamali, and Michael Wallner. 2019. “De la probabilité de creuser un tunnel.” in *21èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications (ALGOTEL).* Saint-Laurent-de-la-Cabrerisse, France

2. Bouillard, Anne, Céline Comte, Élie de Panafieu, and Fabien Mathieu. 2019. “0 = 0, c'est le truc du noyau! Application aux files d'attente.” in *21èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications (ALGOTEL)*. Saint-Laurent-de-la-Cabrerisse, France

Preprints under review

1. de Panafieu, Élie. 2024. “Asymptotic Expansion of Regular and Connected Regular Graphs”. https://arxiv.org/abs/2408.12459
2. Gösgens, Martijn, Lukas Lüchtrath, Elena Magnanini, Marc Noy, and Élie de Panafieu. 2024. “The Erdős-Rényi Random Graph Conditioned on Every Component Being a Clique”. https://arxiv.org/abs/2405.13454

PhD thesis

1. de Panafieu, Élie. 2014. “Analytic Combinatorics of Graphs, Hypergraphs and Inhomogeneous Graphs.”. PhD thesis

Patents

1. **Positioning approach.** Chung Shue Chen, Paolo Baracca, Élie de Panafieu, Diomidis Michalopoulos. Applicant Nokia Technologies Oy. Priority application GB 2310456.5 filed 7 July 2023; published in January 2025 as GB 2631540 A, EP 4488708 A1 and US 2025/0016725 A1. Under examination.

2. **Clustering of a set of items.** Élie de Panafieu, Maria Laura Maag, Quentin Lutz. European patent application EP 3 923 191 A1 (application no. 20179532.5), applicant Nokia Solutions and Networks Oy. Filed 11 June 2020, published 15 December 2021, subsequently withdrawn.